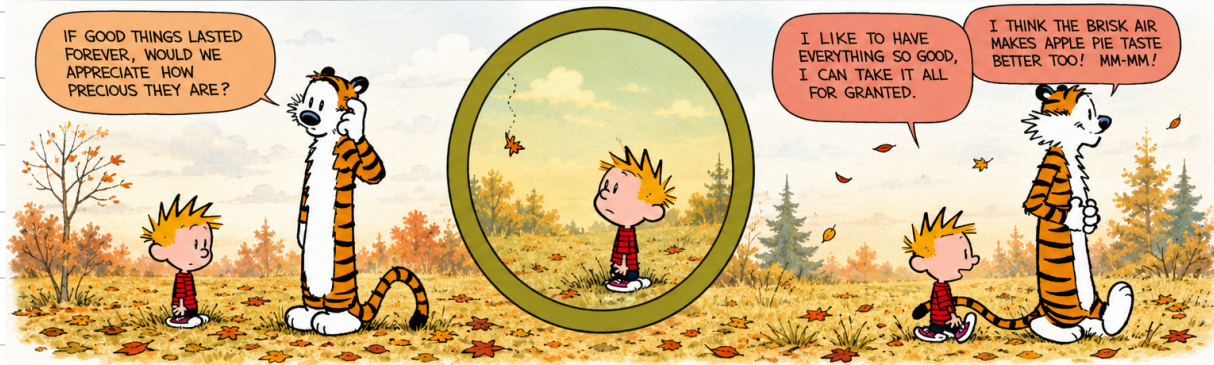


calvin and hobbes

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PRESS SYNDICATE



Agenda

1. BOW
2. Tf-IDF
3. Embeddings
4. Word2Vec
5. CBOW - Continuous Bag of Words
6. Skip-GRAM

Bow

Vocab = { 3

① Cat set

5000 - 80-90

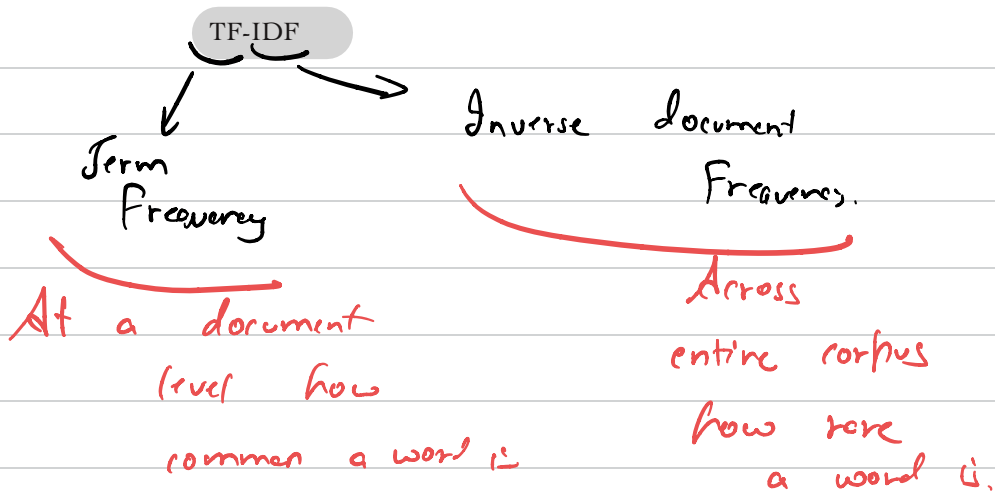
① Apple Bag Cat set _ _ _
0 0 1 1 0 _

Problems

- ① Spelling mistake
- ② Languages → Multiple lang
- ③ Order
- ④ Importance
- ⑤ Good & awesome & bad
- ⑥ Spare

Tweets: 10k tweets.

all 10k tweets comprises of corpus, each tweet independently represents a document.



TF = $\frac{\text{\# of times a 'w' appears in document 'd'}}{\text{Total no of terms in document 'd'}}$

$$IDF = \ln \left(\frac{N \text{ (Total \# of Document)}}{1 + d \text{ (no of doc in which w occurs)}} \right) + 1$$

Example: "I love eating pizza, but pizza is so unhealthy"

Scenario 1:

$$N = 200$$

$$d = 1$$

$$\ln \left(\frac{200}{1+1} \right) + 1$$

$$= 5.6$$

Scenario 2:

$$N = 200$$

$$d = 40$$

$$= \ln \left(\frac{200}{40+1} \right) + 1$$

$$= 2.58$$

Case 1: High TF, very high IDF

This term occurs frequently in the current document, and it occurs rarely across corpus.

This term is probably very important for this document.

Case 2: High TF, Low IDF

This term occurs very frequently in current document, but occurs mostly in all document.

Probably some generic word, like "AI", or "The".

Case 3: Low TF, Low IDF

The word occurs rarely in current document, but occurs a lot in most document.

Probably not very important

Case 4: Low TF, High IDF

This word occurs rarely in current document, but it's quite rare also in most documents. Could be important.

Embeddings!!

Embeddings are 1d representation of a word, that 1D representation basically represents the "gist" of the word.

King = []₁₀₀

Queen = []₁₀₀



King - Man + Woman = Queen

Royalty

Passage A

1. After storing enough acorns, the squirrel curled up in its nest to _____ through the entire winter.
2. Many reptiles slow their heartbeat to almost nothing when they _____ in cold caves.
3. Because animals _____ for months, they must build up fat reserves beforehand.

Passage B

When Mia opened the letter, she gasped and looked 驚いた ; she had never expected to win the scholarship

Tokenization

Machine learning is deep
I love fun

4, 1, 6, 5
2, 3, 7

Word → Index mapping:

```
'learning' → 1  
'i' → 2  
'love' → 3  
'machine' → 4  
'deep' → 5  
'is' → 6  
'fun' → 7  
'amazing' → 8
```

Indiàment → Indian Amazon

Diamonds

Gold bars

Gucci

→

Gucci

Products $\rightarrow$ 100 words in the title

90% $\rightarrow$ 20 words, 10% $\rightarrow$ 100 words

95th percentile $\rightarrow$ mean

List = [6, 5, 10, ..., ...]
95th percentile $\rightarrow$ mean.

10 million rows of data, some of the rows had about 100 words in them, most rows about 20 words.

RAM should have capacity to load 10 million rows X 100

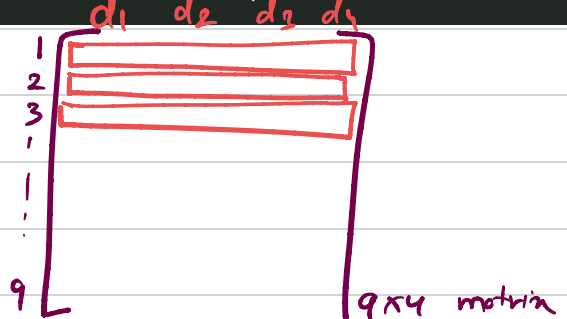
10 million rows X 20

Word $\rightarrow$ Index mapping:

'learning' $\rightarrow$ 1 $\rightarrow$ [1000
'i' $\rightarrow$ 2 $\rightarrow$ [1000
'love' $\rightarrow$ 3 $\rightarrow$ [1000
'machine' $\rightarrow$ 4
'deep' $\rightarrow$ 5
'is' $\rightarrow$ 6
'fun' $\rightarrow$ 7
'amazing' $\rightarrow$ 8

EMBEDDING LAYER CONFIGURATION

- Vocabulary size (input_dim): 9
- Embedding dimension (output_dim): 4
- Input sequence length: 5
- Embedding matrix shape: (9, 4)
 $\rightarrow$ 9 words $\times$ 4 dimensions
 $\rightarrow$ Total trainable parameters: 36



Caring Low gender responsible rich.
 $Shawi = [9 \ 9 \ 0 \ 1 \ 1]$

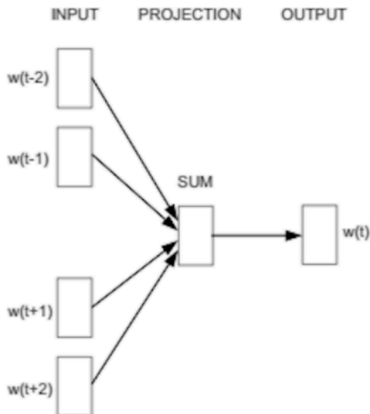
Pad $\rightarrow$ [Embedding] $\rightarrow$ GPU

Bow

TF-IDF

Embedding $\rightarrow$ tokenize $\rightarrow$ Convert text to tokens
 $\rightarrow$ Pad Sequences
 $\rightarrow$ Embedding (Sequences)
 $=$ Embeddings

Continuous Bag of Words



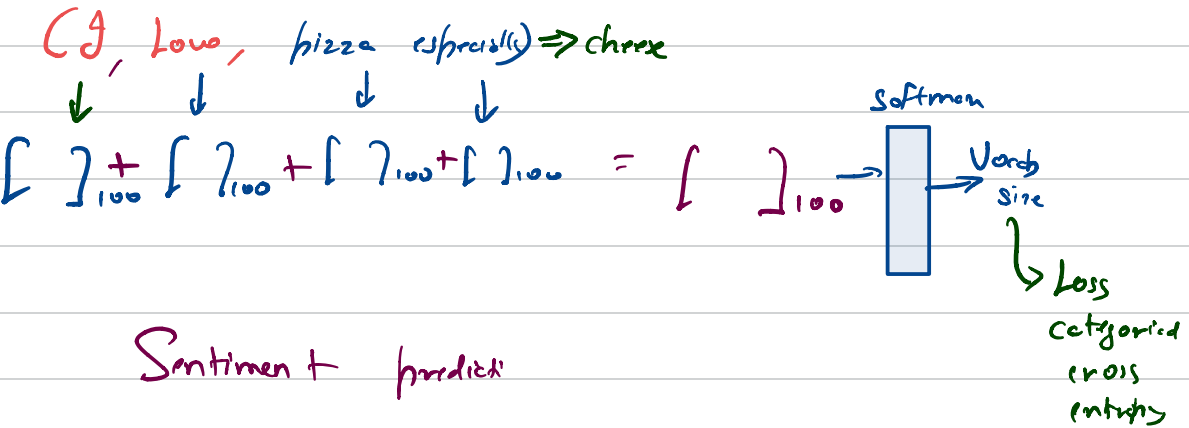
I love cheese pizza especially cheese burst.

window = 2

CBow

- ① (I, love, pizza especially) $\Rightarrow$ cheese
- ② (love, cheese, especially, cheese) = pizza.

Embedder = 100



This movie is awesome

$\downarrow \quad \downarrow \quad \downarrow \quad \downarrow$

$\lfloor \rfloor_{100} + \lfloor \rfloor_{100} + \lfloor \rfloor_{100} + \lfloor \rfloor_{100}$

Should I choose 10, 100, 1000, 10000, 100000

spouse in 10 words?
100 words?
1 lakh words??

$$\begin{matrix} \square & \times & \square \\ -1/1 & & 2/3 \end{matrix}$$



Your salary is \$100,000

This number is a million

↓ ↓ ↓ ↓

$$\underbrace{1,000 + 1,000 + 1,000 + 1,000}_{4} = 1,000 \rightarrow \boxed{} \rightarrow 0/1$$

4
 Dog bits Men
 Men bits Dog

5000

5 → 51

